

Chapter 1. Introduction

Surface science is the study of physical and chemical phenomena that occur at the interface of two phases, including solid–liquid interfaces, solid–gas interfaces, solid–vacuum interfaces, and liquid–gas interfaces. It includes the fields of surface chemistry and surface physics.

We will focus on those aspects that are related to thin films, including their fabrication/preparation, structures, properties and applications.

Thin films are deposited onto bulk materials (substrates) to achieve properties unattainable or not easily attainable by substrates alone.

How thin?

~0.5 nm – 1 mm, a difference of 6 orders of magnitude

Surface science and thin film technology are closely related. They overlap largely with each other in content. We therefore pack them together in one course.

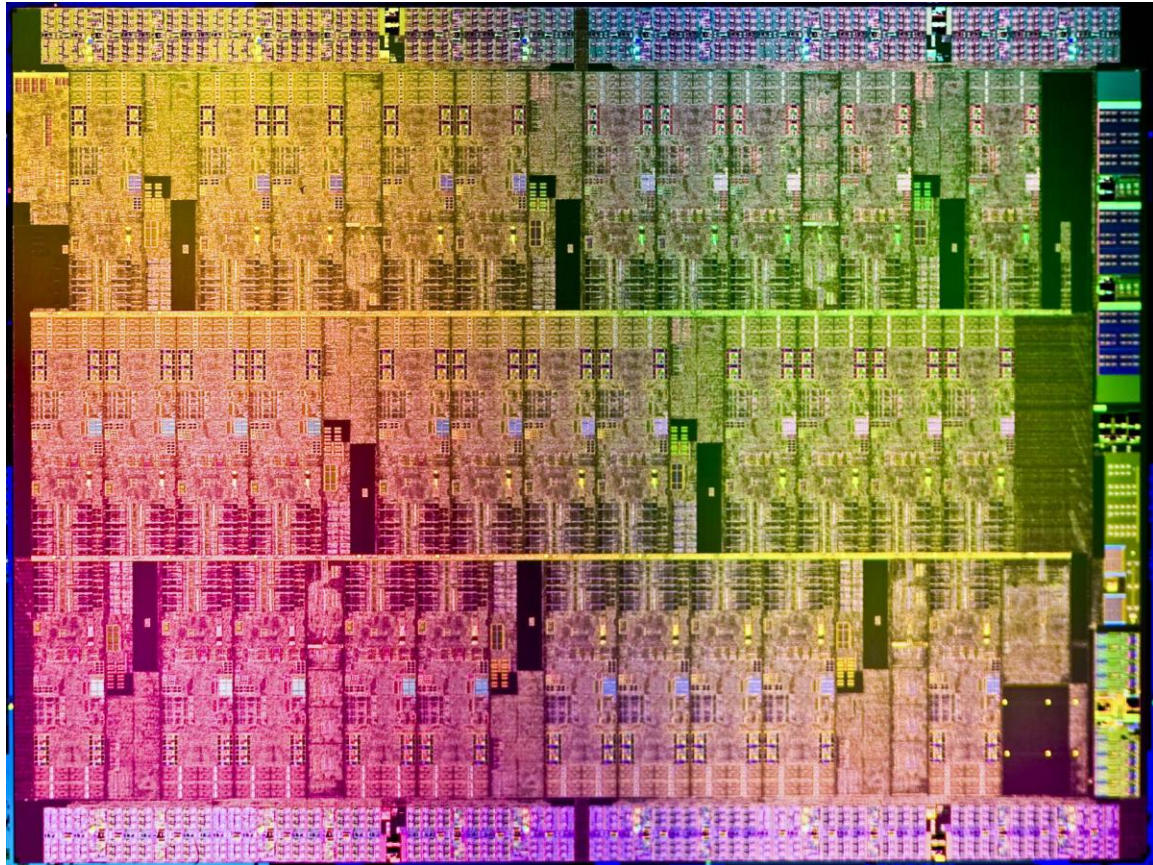
Many existent technologies are more or less based on thin films.

Properties

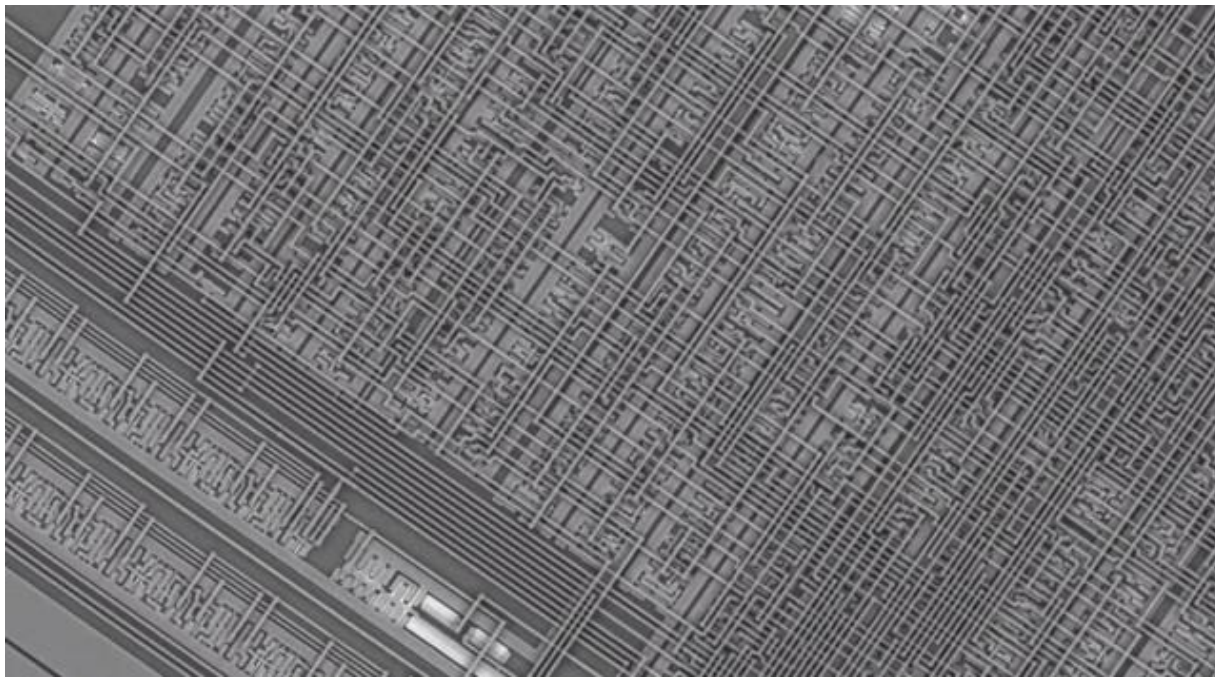
Applications

Optical	Reflective/antireflective coatings; interference filters; decoration (color, luster); waveguides; displays; solar cells; photodetector arrays
Electrical/ electronic	Insulation; conduction; semiconductor devices; piezoelectric drives; integrated circuits; solar cells; displays; photodetector arrays
Magnetic	Memory disks
Chemical	Barriers to diffusion or alloying; protection against oxidation or corrosion; gas/liquid sensors
Mechanical	Tribological (wear-resistant) coatings; hardness; adhesion; micromechanics
Thermal	Barrier layers; heat sinks

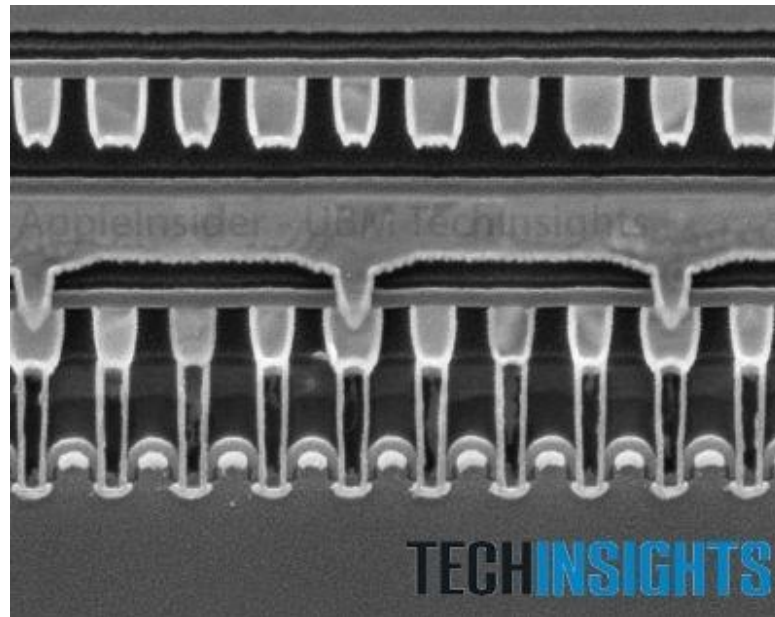
Integrated Electronic Circuits



Inside top

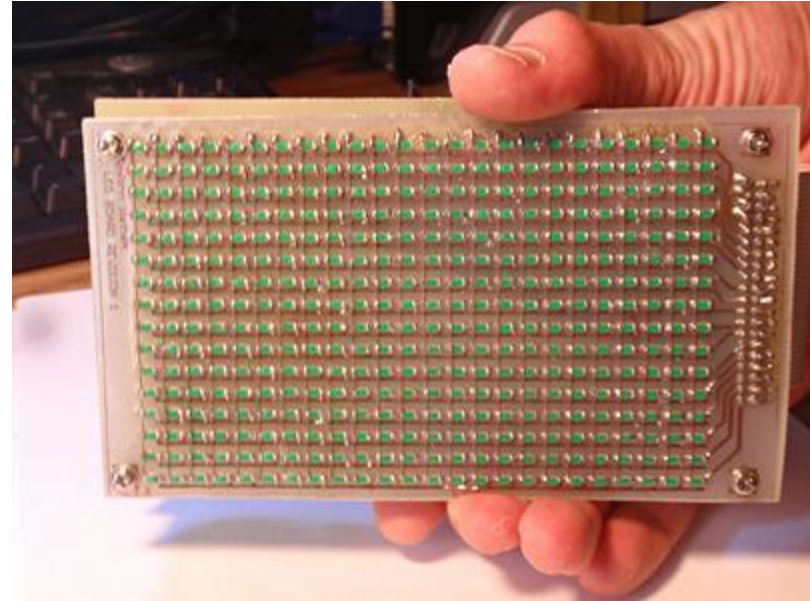


Inside
cross-section

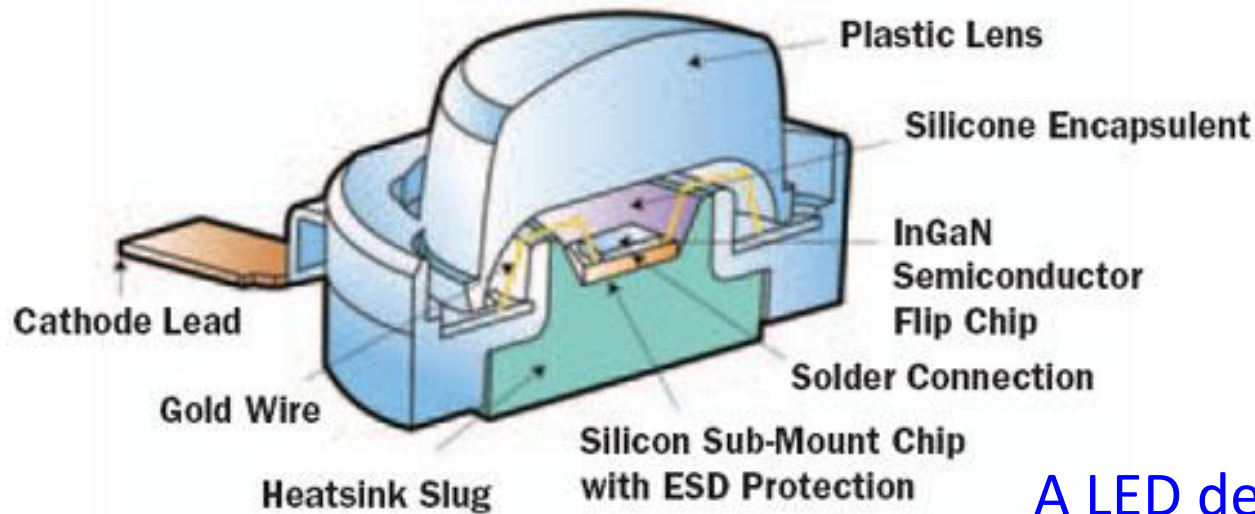


SRAM transistors in Apple's iPad 2 A5

LED Arrays as Displays

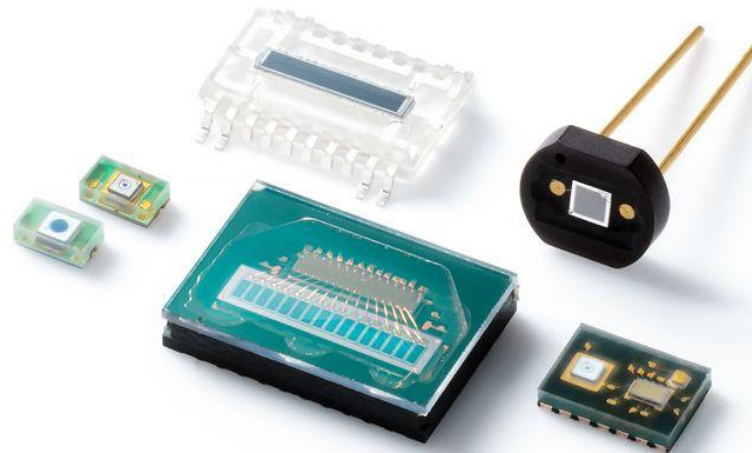
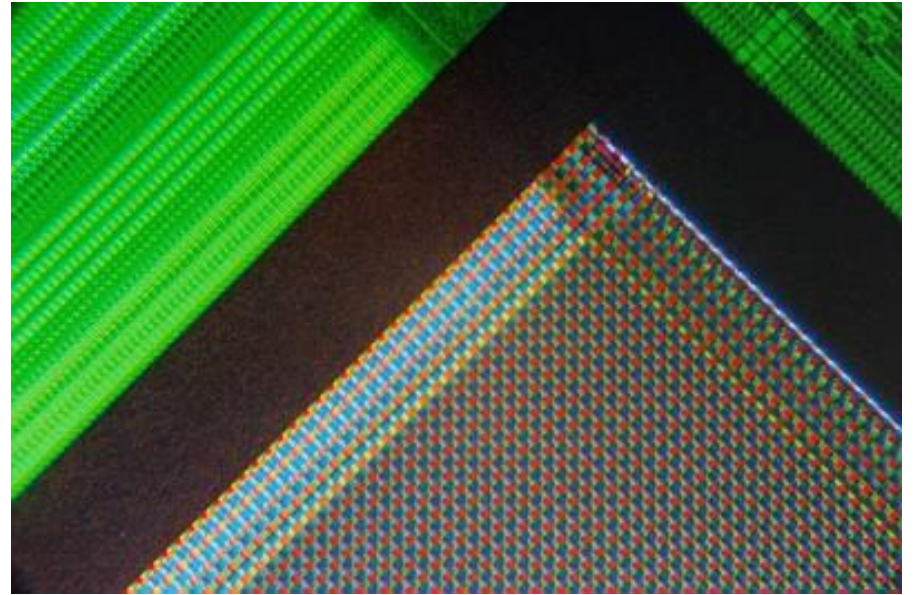


LED matrix



A LED device

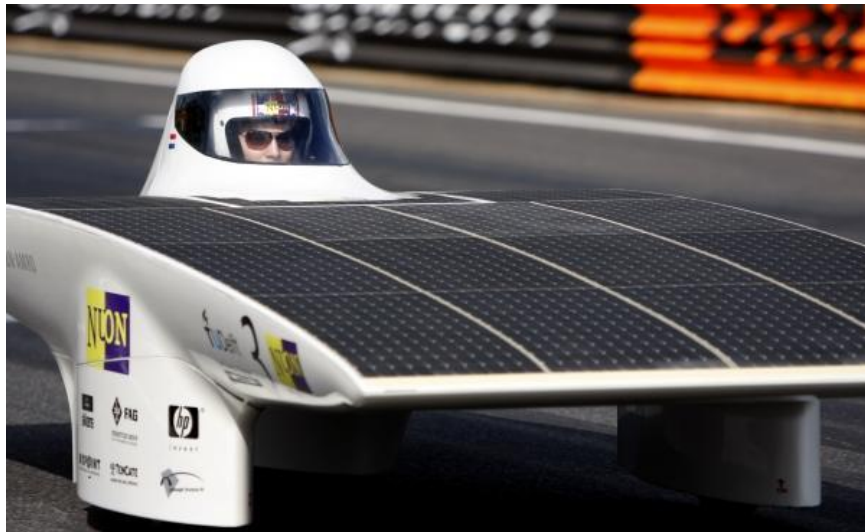
Photodetector Arrays in Digital Cameras



Solar Cell Plant



Cars and Airplanes Powered by Photovoltaic Devices



Optical Mirrors



Broadband Dielectric Mirrors



Broadband Metallic Mirrors



Laser Line Mirrors



Ultrafast Mirrors



Retroreflectors



Parabolic Mirrors



Concave Mirrors



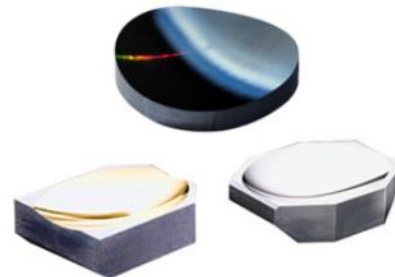
Elliptical Mirrors



Square Mirrors



D-Shaped Mirrors



Custom Freeform Mirrors

Optical Filters



Bandpass Filters



Longpass Filters



Shortpass Filters



Neutral Density Filters



Color and Absorptive Optical Filters



Dichroic Filters



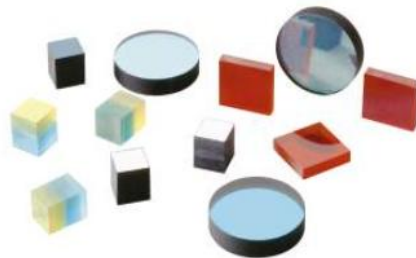
Fluorescence Optical Filters



Diffusers

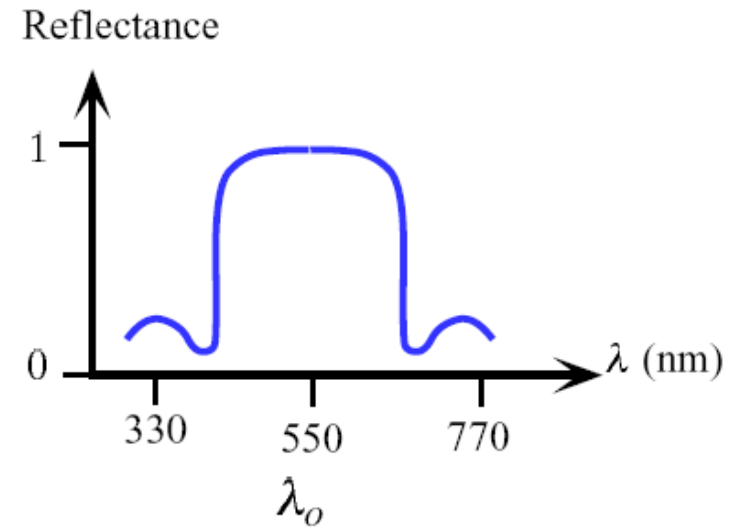
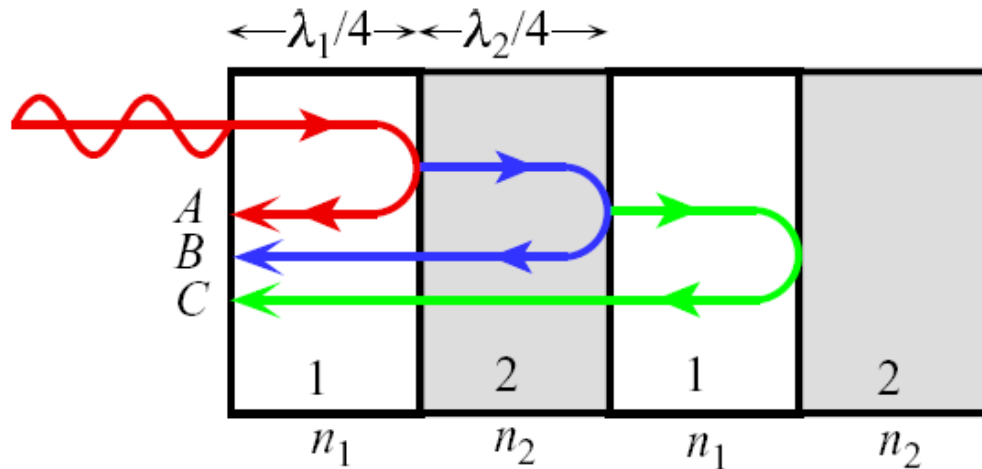


Optical Filter Sets



Custom Optical Filters

Multilayer Films of Different Dielectric Materials



Dielectric mirrors consist of multiple layers of films alternating between high and low indexes of refraction. A, B, and C waves are in phase with each other and interfere constructively.



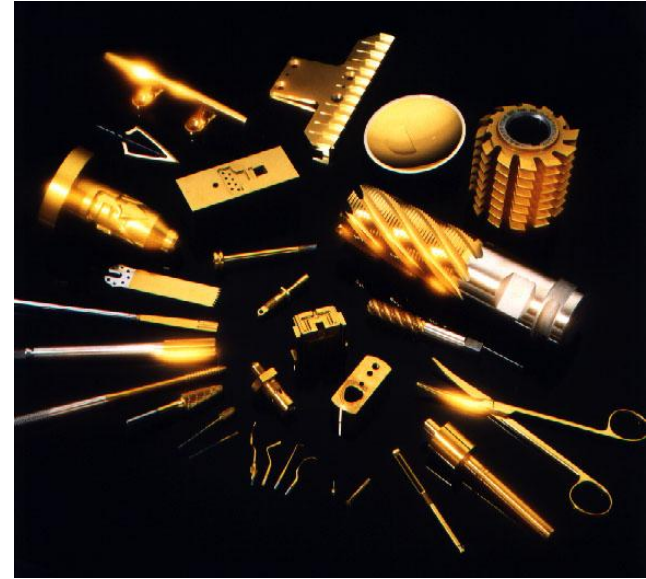
The manufacture of commercial electronic, photonic, and optoelectronic devices is dominated by thin film technologies.

Thin Films for Mechanical Properties



Cr coatings on automobile parts

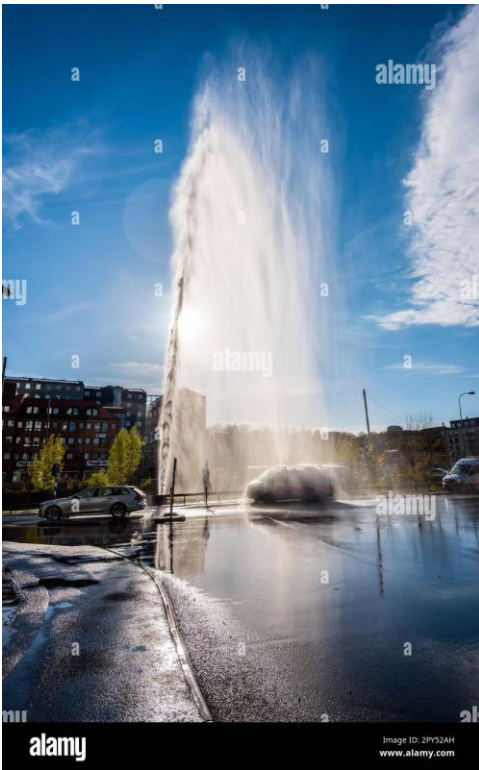
Impart hardness, metallic luster, and protection against ultraviolet light. Cr coatings on plastic parts achieve the same functionality as the bulk metal but at significant savings in cost and weight.



TiN coatings on cutting tools

Offer hardness, low friction, and a chemical barrier to alloying of the tool with workpieces.

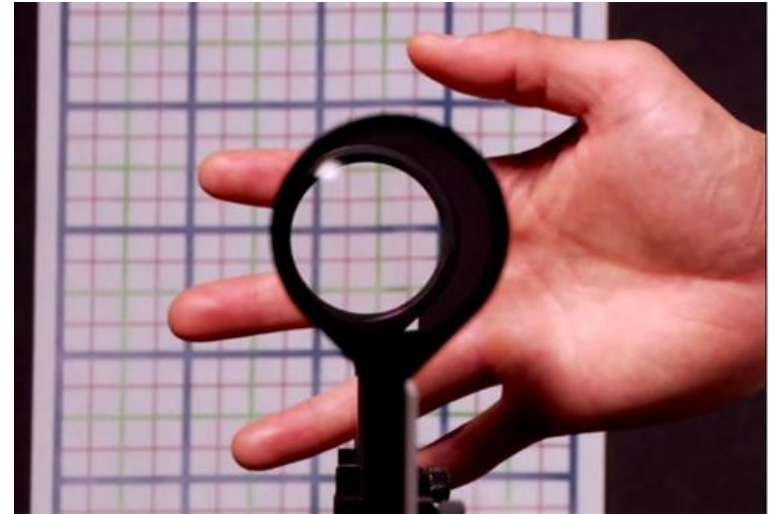
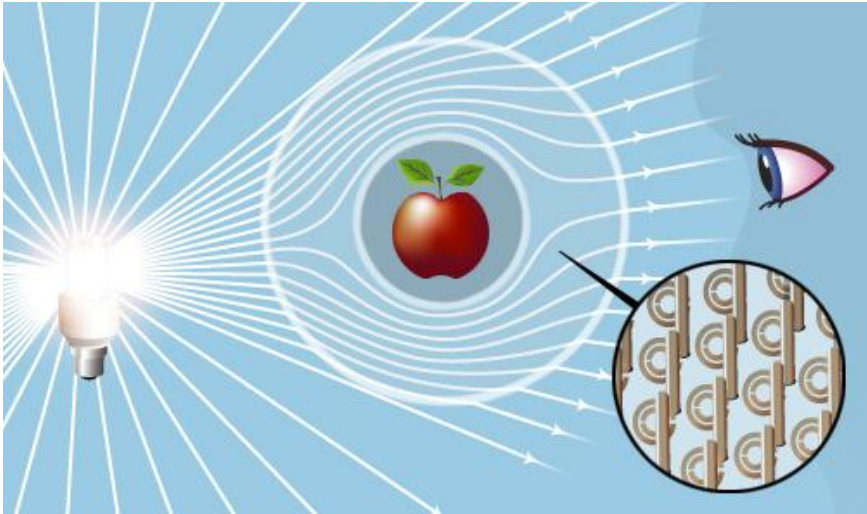
Coatings for Prevention from Corrosion



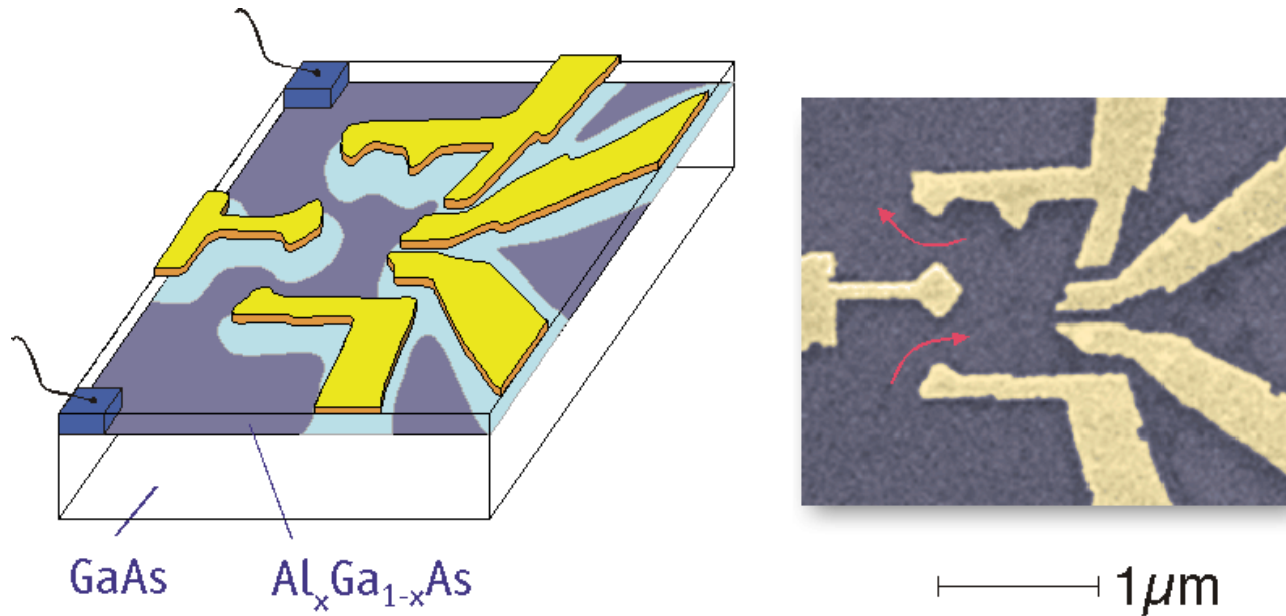
Stealth Aircraft: employing a combination of technologies to reduce visibility in the visible, infrared (700 nm – 1 mm), radio (3 kHz – 300 GHz, for radar), and audio (20 Hz – 20 kHz) frequency spectral regions.



Cloaking in the Visible Region



Semiconductor Heterostructures



2D electron gas: Electrons are confined vertically to the ground state of a quantum well located at a GaAs/AlGaAs interface, and form a 2D electron gas (2DEG). A mean free path and coherence length on the order of $10\ \mu\text{m}$ insure that the carriers are coherent and ballistic. Metallic gates deposited by electron beam lithography confine electrons laterally. A number of interesting phenomena related to [the quantum Hall effect \(integer, 1985 Nobel Prize and fractional, 1998 Nobel Prize\)](#) have been discovered on this system.

Two More Examples Closer to Current Research

- ✿ Diamond films and wafers
- ✿ Metallic quantum wells



QMWG800 Synthetic Diamond HPHT Hydraulic Cubic Press

Luoyang, Henan, China

Diamond

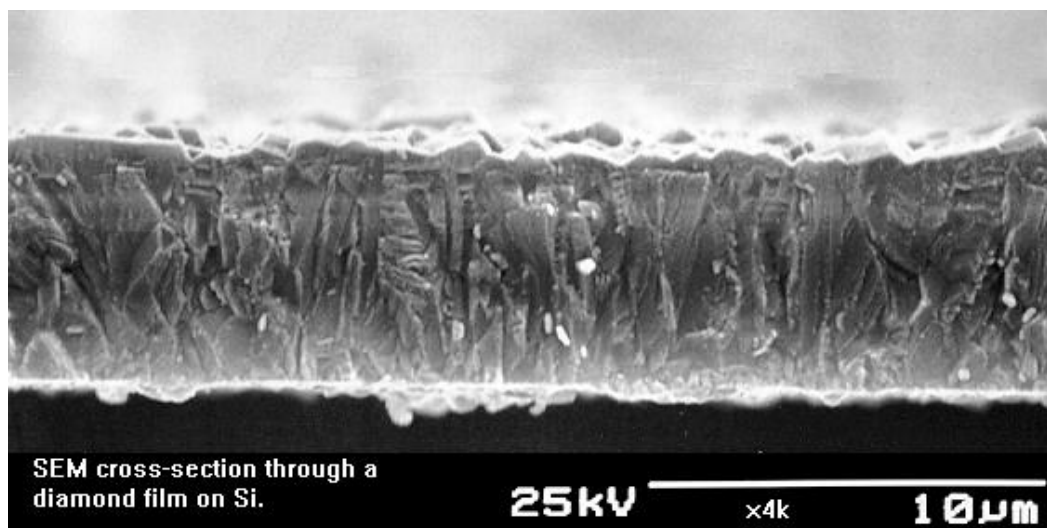
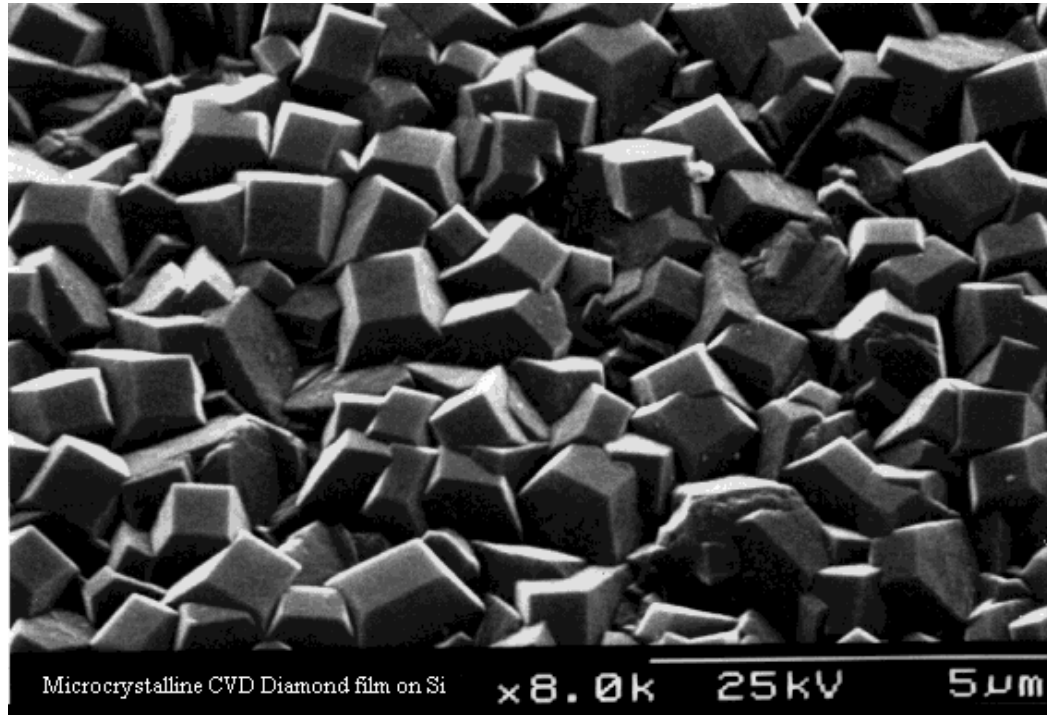


In addition to its unmatched beauty, diamond has a number of other remarkable properties:

- (1) Hardest substance ever known ($H_v > 8000 \text{ kg mm}^{-2}$);
- (2) Higher elasticity modulus ($E = 1050 \text{ GPa}$) than any other material;
- (3) One of the highest resistivities ($\rho > 10^{16} \Omega \text{ cm}$) when no impurities are present;
- (4) Very high thermal conductivity ($\kappa = 1100 \text{ W m}^{-1} \text{ K}^{-1}$) exceeding those of Cu and Ag;
- (5) Low thermal expansion coefficient ($\alpha = 1.2 \times 10^{-6} \text{ K}^{-1}$) to yield high resistance to thermal shock;
- (6) Resistance to chemical attack.

These facts have spurred exciting and competitive quests in materials science: the preparation of diamond. The most common method for depositing diamond films is chemical vapor deposition (CVD).

Diamond Films



Application	Properties Required	Comments
1. Resonator diaphragms of tweeter loud speakers	High modulus of elasticity	Frequency response up to 60,000 Hz possible; commercially available
2. Ultrahard tool coatings	High hardness	Commercially available
3. Sunglass lenses	High hardness, scratch resistance, optical transparency	Commercially available
4. Computer hard disk coatings	High hardness, low wear	Coatings minimize head-disk contact wear
5. Watch cases	High hardness, scratch resistance	
6. Prosthetic devices	High hardness, low wear	
7. Optical coatings	High hardness, high index of refraction	
8. Infrared laser window	Transparency to IR	
9. Electronic devices—traveling wave amplifiers		Heteroepitaxial films required
10. Semiconductor device heat sinks	High thermal conductivity	Commercially available
11. High-temperature semiconductor devices	Large energy band gap	
12. Abrasive grain	High hardness	

Nonstick pan



Medical implants



Piston



Part in AK47 rifle

Diamond Wafers

Diamond can also be an outstanding semiconductor material, superior in many ways to silicon. Diamond devices could operate at higher temperatures ($>400\text{ }^{\circ}\text{C}$) and higher power than those of silicon, as well as being faster, denser and more resistant to radiation. But practical diamond electronics will need large-area, single-crystalline diamond wafers to be fabricated, analogous to silicon wafers commonly used in the semiconductor industry.

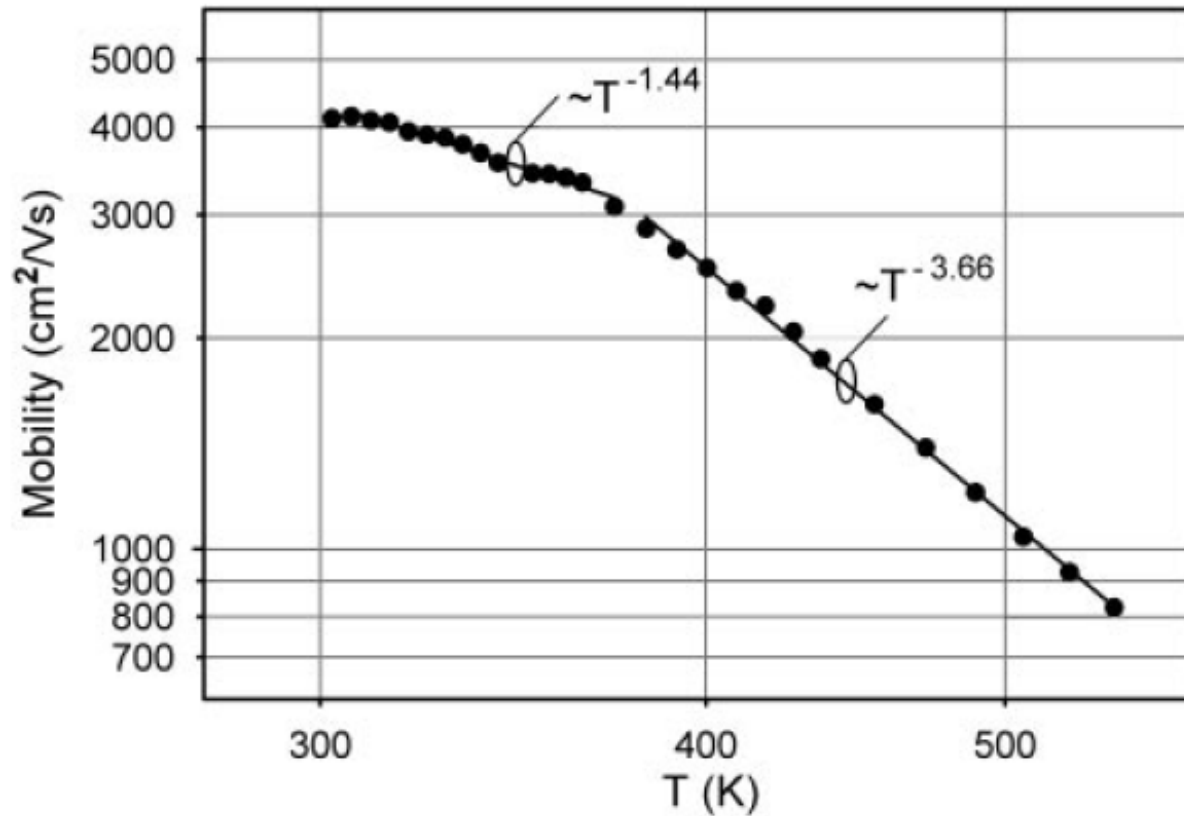
Diamond can be grown on diamond (**homoepitaxy**) by chemical vapor deposition. Electronics-grade diamond has been made in this way. But single-crystal diamond substrates are small and expensive.

The alternative is to grow diamond on a foreign (non-diamond) single-crystal wafer (**heteroepitaxy**).

Comparison between Diamond and Silicon

	Si	4H-SiC	GaN	Natural Diamond	CVD Diamond	Potential device application benefit
Bandgap (eV)	1.1	3.2	3.44	5.47	5.47	High temperature
Breakdown field (MVcm ⁻¹)	0.3	3	5	10	10	High voltage
Electron saturation velocity (x10 ⁷ cm s ⁻¹)	0.86	3	2.5	2	2	High frequency
Hole saturation velocity (x10 ⁷ cm s ⁻¹)	n/a	n/a	n/a	0.8	0.8	
Electron mobility (cm ² V ⁻¹ s ⁻¹)	1450	900	440	200–2800	4500	
Hole mobility (cm ² V ⁻¹ s ⁻¹)	480	120	200	1800–2100	3800	
Thermal conductivity (Wcm ⁻¹ K ⁻¹)	1.5	5	1.3	22	24	High power
Johnson's figure of merit	1	410	280	8200	8200	Power-frequency product
Keyes' figure of merit	1	5.1	1.8	32	32	Transistor behavior thermal limit
Baliga's figure of merit	1	290	910	882	17200	Unipolar HF device performance

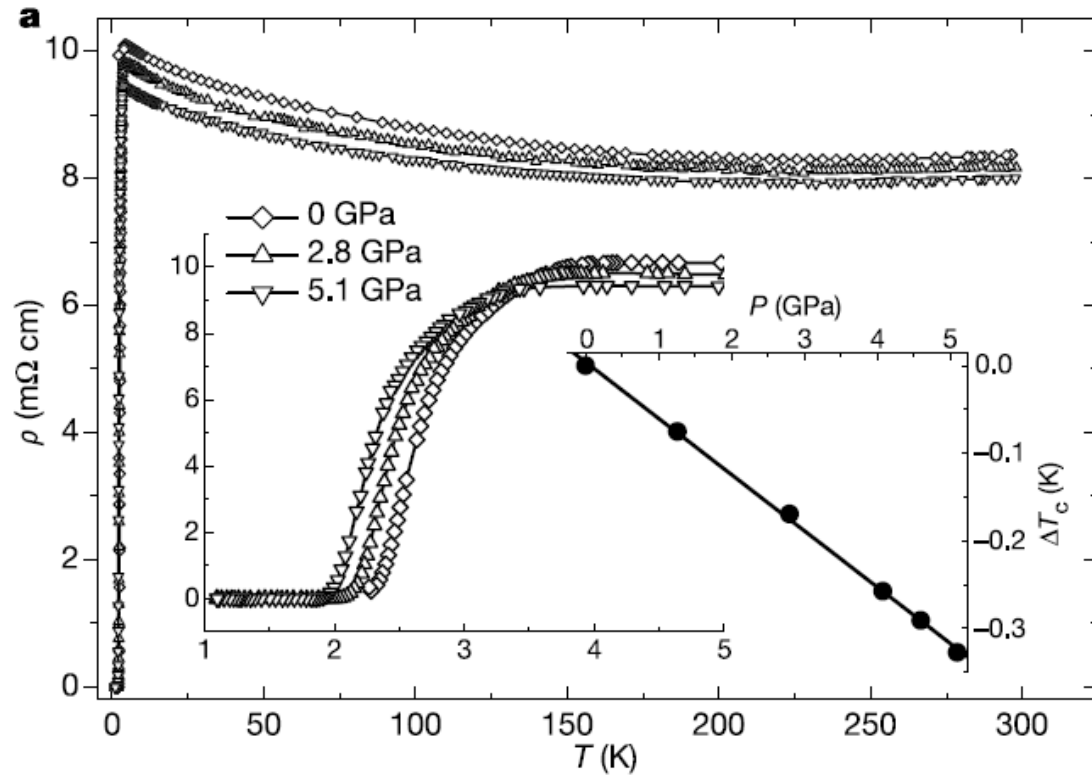
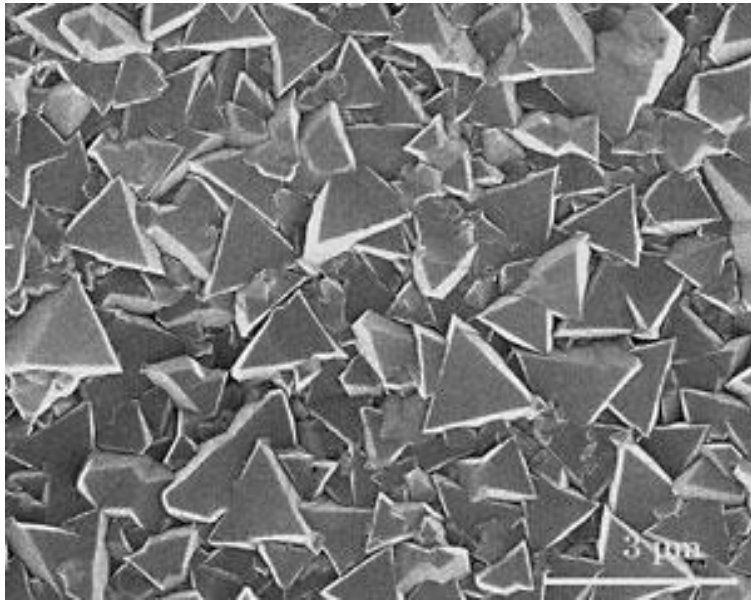
Room-temperature drift mobilities of $4500 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ for electrons and $3800 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ for holes have been measured in high-purity single-crystalline diamond. Such high charge mobilities of single-crystalline diamond films are encouraging for the development of high-performance diamond electronics.



J. Isberg, J. Hammersberg, E. Johansson, T. Wikström, D. J. Twitchen, A. J. Whitehead, S. E. Coe, G. A. Scarsbrook, *Science* **2002**, 297, 1670.

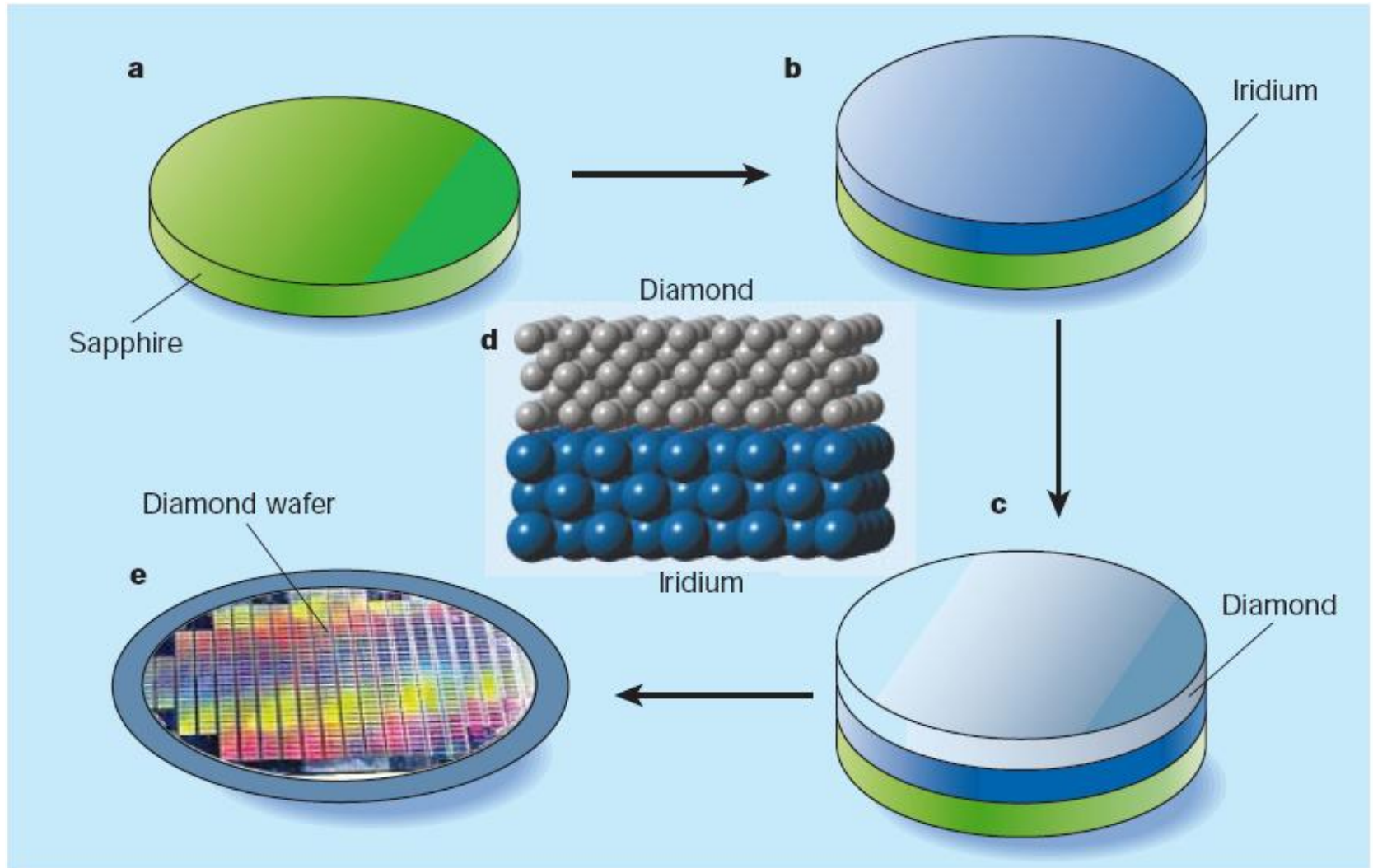
Superconductivity in Diamond

Boron is relatively easily doped into diamond as a charge acceptor. Superconductivity has been discovered in B-doped diamond synthesized at high pressure (nearly 10^5 atm) and temperature (2500–2800 K). The superconducting transition temperature T_c is ~ 4 K.

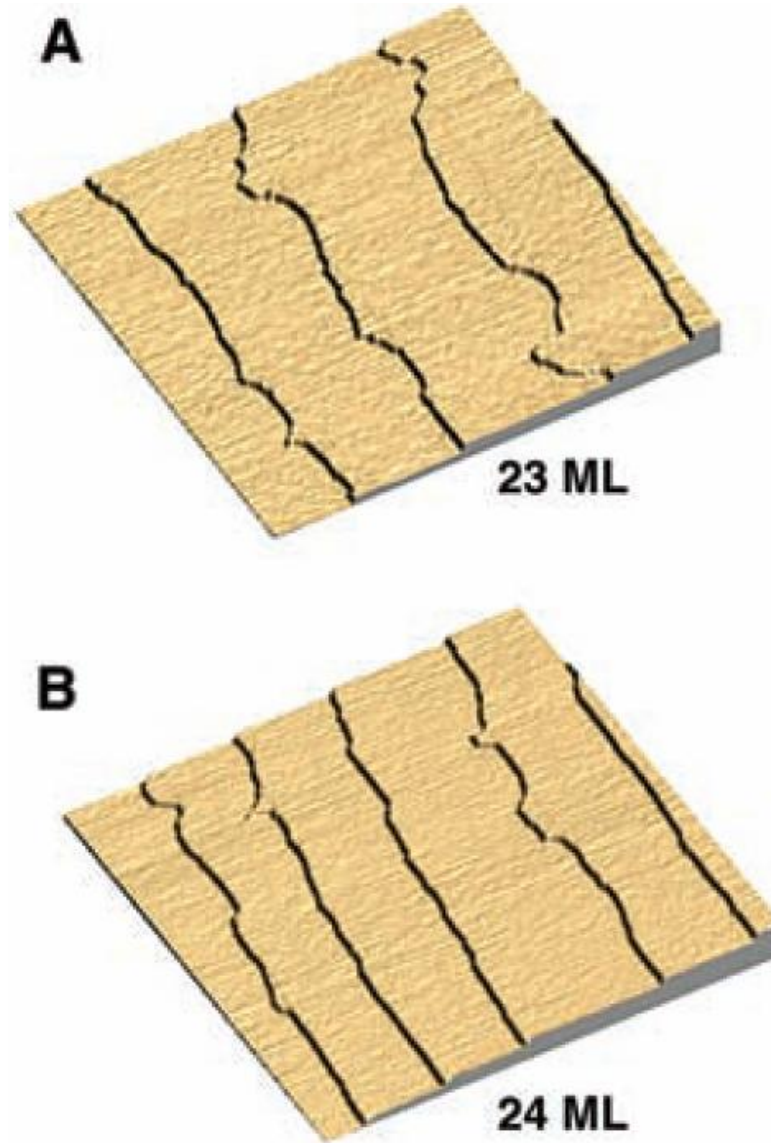


E. A. Ekimov, V. A. Sidorov, E. D. Bauer, N. N. Mel'nik, N. J. Curro, J. D. Thompson, S. M. Stishov, *Nature* **2004**, 428, 542.

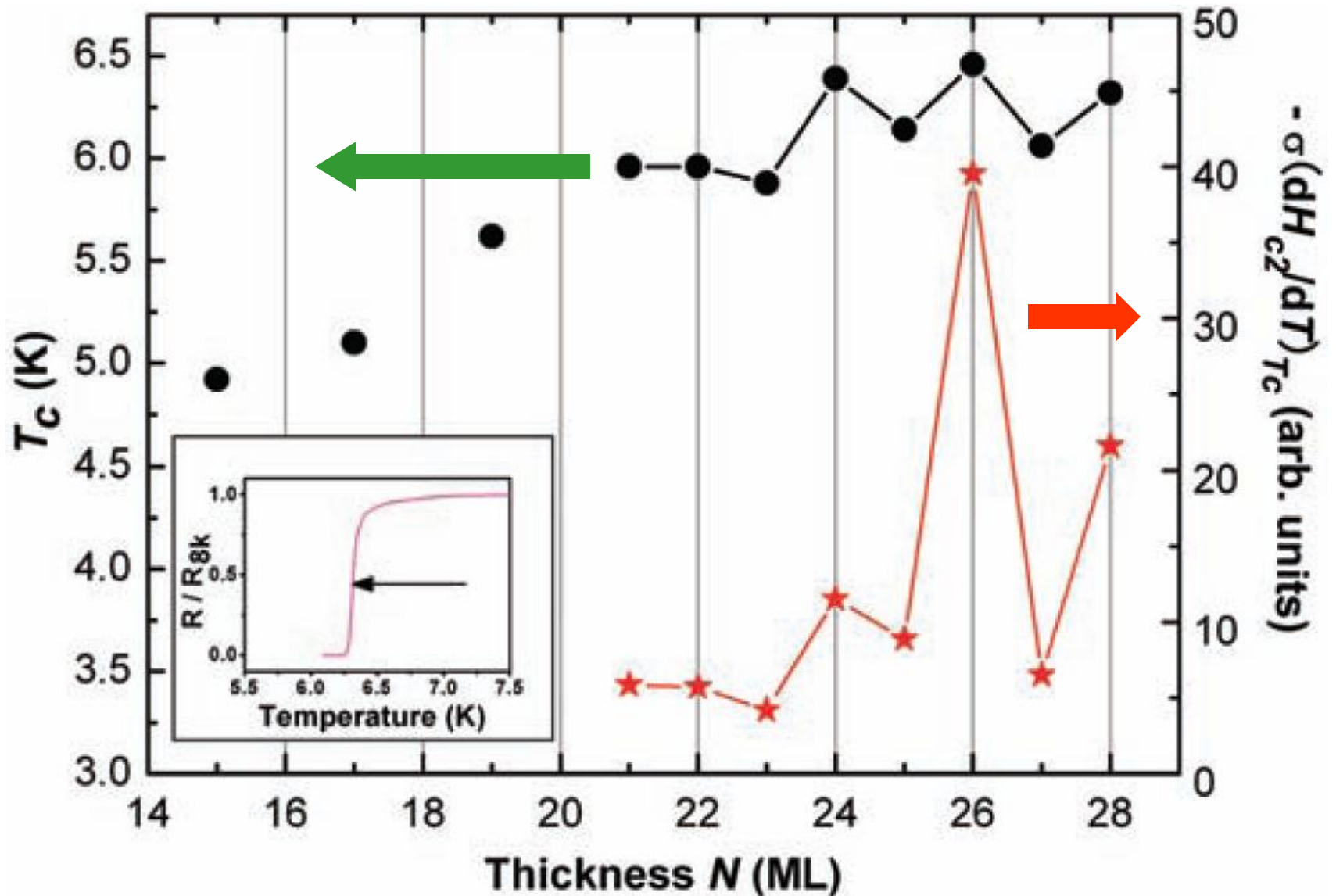
Epitaxial Growth



Pb Films Grown on Si Substrates



Superconducting T_c Changes



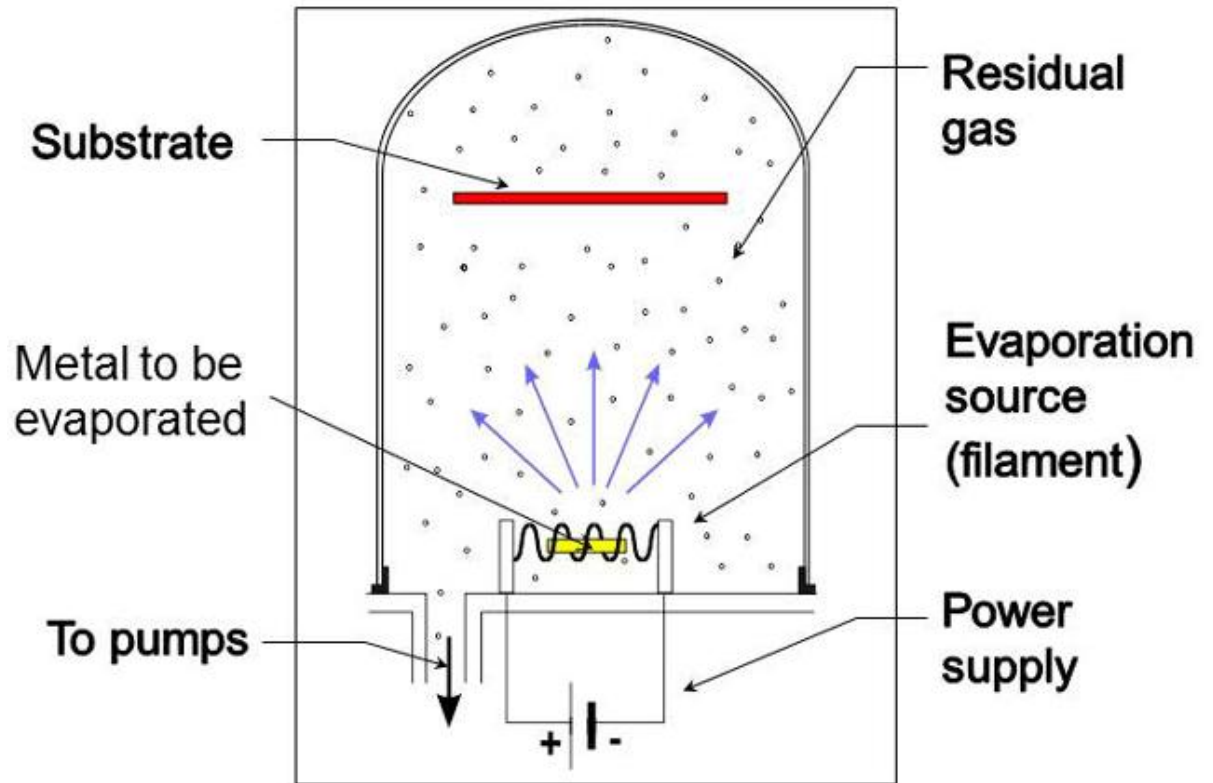
This course will focus on the deposition of thin films from the vapor phase.

We will not discuss the deposition of thin films from the liquid phase or films grown by reaction with the substrate rather than by deposition, such as SiO_2 grown by the thermal reaction of water vapor with Si.

Vapor phase thin-film techniques have three advantages: (1) applicability to any material, (2) wide adjustability in substrate temperature, and (3) access to the surface during deposition.

Liquid-phase techniques, such as electroplating, are useful for low-cost production. They will not be covered in this course.

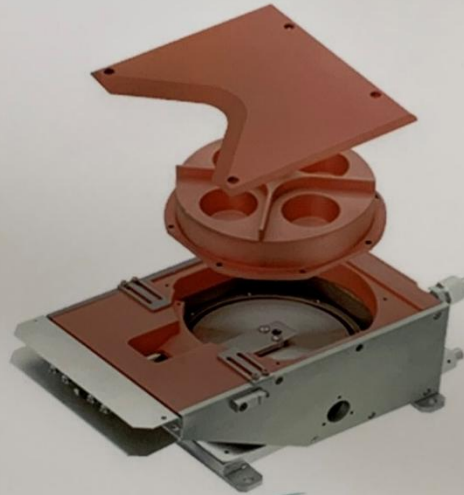
A typical Thermal Evaporation System



Our Electron-Beam Deposition System



Junsun Tech EBS-500
E-beam Deposition System



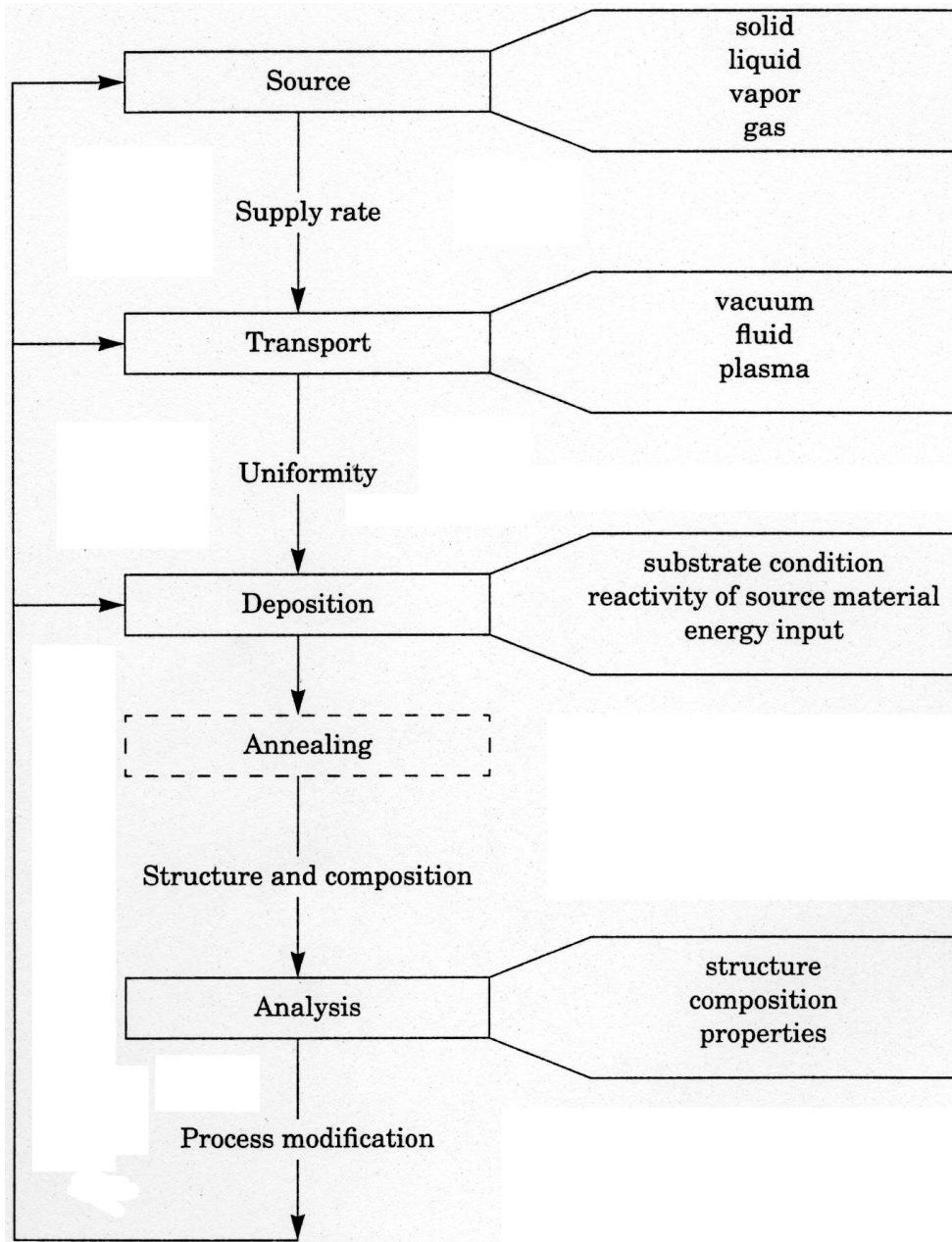
Temescal E-gun with 4 pockets



Gold thin film deposited on Si substrate

Venue: Room G24, Science Center North Block

Process Steps



Solid materials are vaporized by heat or by an energetic beam of electrons, photons (laser ablation), or positively ions (sputtering) in order to transport them to substrates. These methods are categorized as **physical vapor deposition (PVD)**. Thin-film processes that use gases, evaporating liquids, or chemically gasified solids as source materials are categorized as **chemical vapor deposition (CVD)**.

The key difference is that PVD is a physical transfer from solid-to-solid (through vapor), while CVD is a chemical reaction from gas-to-solid (often with flow control).

A major issue in the transport step is the uniformity of the arrival rate over the substrate. In a **high vacuum**, molecules travel from the source to the substrate in straight lines, whereas in a **fluid** there are many molecular collisions during the transport step. Therefore, in a **high vacuum**, the uniformity is determined by the geometry, whereas in a **fluid** it is determined by gas flow patterns and by the diffusion of source molecules.

Substrate surface conditions include roughness, contamination level, degree of chemical bonding with arriving materials, and crystallographic parameters.

The reactivity factor refers to the probability of arriving molecules reacting with the surface and becoming incorporated into the film. The probability is known as the “**sticking coefficient**”, S_c , which can vary from 1.0 to $<10^{-3}$.

Energy input, including substrate heating, energetic electrons, photons, plasma, chemical reactions, has profound effects on both the reactivity of arriving materials and on the composition and structure of films.

In physics, a fluid is any substance that continuously deforms or flows when a shear stress is applied. It includes liquids, gases, and even plasmas.

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